```
template<typename Callable>
packaged_task(Callable&& func);
```

Preconditions

The expression `func(args...)` shall be valid, where each element `args-`*i* in `args...` shall be a value of the corresponding type `ArgTypes-`*i* in `ArgTypes...`. The return value shall be convertible to `ResultType`.

Effects

Constructs a `std::packaged_task` instance with an associated asynchronous result of type `ResultType` that isn't *ready* and an associated task of type `Callable` that's a copy of `func`.

Throws

An exception of type `std::bad_alloc` if the constructor is unable to allocate memory for the asynchronous result. Any exception thrown by the copy or move constructor of `Callable`.

Std::Packaged_Task Construction From A Callable Object With An Allocator

Constructs a `std::packaged_task` object with an associated task and asynchronous result, using the supplied allocator to allocate memory for the associated asynchronous result and task.

Declaration

```
template<typename Allocator,typename Callable>
packaged_task(
    std::allocator_arg_t, Allocator const& alloc,Callable&& func);
```

Preconditions

The expression `func(args...)` shall be valid, where each element `args-`*i* in `args...` shall be a value of the corresponding type